

MedAssist AI

Turning Clinical Data into Human Understanding

CS 595 — Medical Informatics & AI



Illinois Institute of Technology



Leap of Faith®

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Abstract

MedAssist AI is a full-stack medical informatics platform that transforms blood test reports into comprehensive, patient-friendly health insights. A healthcare patient's lab report — dense with abbreviations, reference ranges, and clinical codes — is opaque to most non-clinical readers. MedAssist bridges this gap by combining Google Gemini Vision OCR, a multi-provider LLM ensemble with consensus judging, and evidence-based clinical risk scoring to convert raw lab values into plain-language summaries, personalized diet and lifestyle plans, FDA-verified medication suggestions, and structured follow-up schedules. Built as the capstone project for CS 595 — Medical Informatics & AI at Illinois Institute of Technology, the platform demonstrates responsible AI design in healthcare: explainable outputs, HIPAA-compliant audit logging, role-based access control, and conservative risk-first decision merging across all AI agents.

1. Introduction

Routine blood tests are among the most information-dense documents a patient receives, yet they are rarely explained in language a non-clinical reader can act on. A CBC panel or metabolic profile can reveal early signs of kidney disease, cardiovascular risk, or nutritional deficiency — but only if the reader understands what the numbers mean. Clinicians rarely have time to walk patients through every value during a short appointment, leaving patients to interpret results on their own.

MedAssist AI addresses this problem by acting as an intelligent intermediary between the raw clinical data and the patient. The system accepts a scanned or photographed blood report, extracts all measurable parameters using computer vision, analyzes them through a multi-LLM ensemble, calculates validated clinical risk scores, and presents everything in plain English — with audio narration, trend charts, and a downloadable PDF summary. The goal is not to replace physicians, but to ensure patients arrive at their next appointment informed, with meaningful questions and a clear picture of their health trajectory.

2. Problem Statement & Motivation

2.1 The Gap in Medical Data Literacy

A 2022 survey by the American Board of Internal Medicine found that over 60% of patients who received lab results could not correctly identify which values were abnormal. When abnormal values are identified, fewer than half of patients understand the clinical significance or know what action to take. This data literacy gap is widest among patients with limited healthcare experience, low health literacy, or language barriers.

2.2 The Scope of MedAssist AI

MedAssist AI focuses on the patient-side of the data literacy problem:

- Convert scanned or photographed blood reports into structured, searchable data
- Explain each abnormal value in plain language with clinical significance

- ### 3. System Architecture — High-Level Design (HLD)

MedAssist AI follows a three-tier architecture: a React 19 SPA on the frontend (Vercel), an Express.js REST API on the backend (Render), and a PostgreSQL database on Supabase. All AI processing runs server-side through a multi-provider LLM ensemble. The frontend communicates with the backend via Axios with a JWT interceptor. Long-running agent jobs execute as fire-and-forget background processes, and the client polls an agent status endpoint to stream progress.

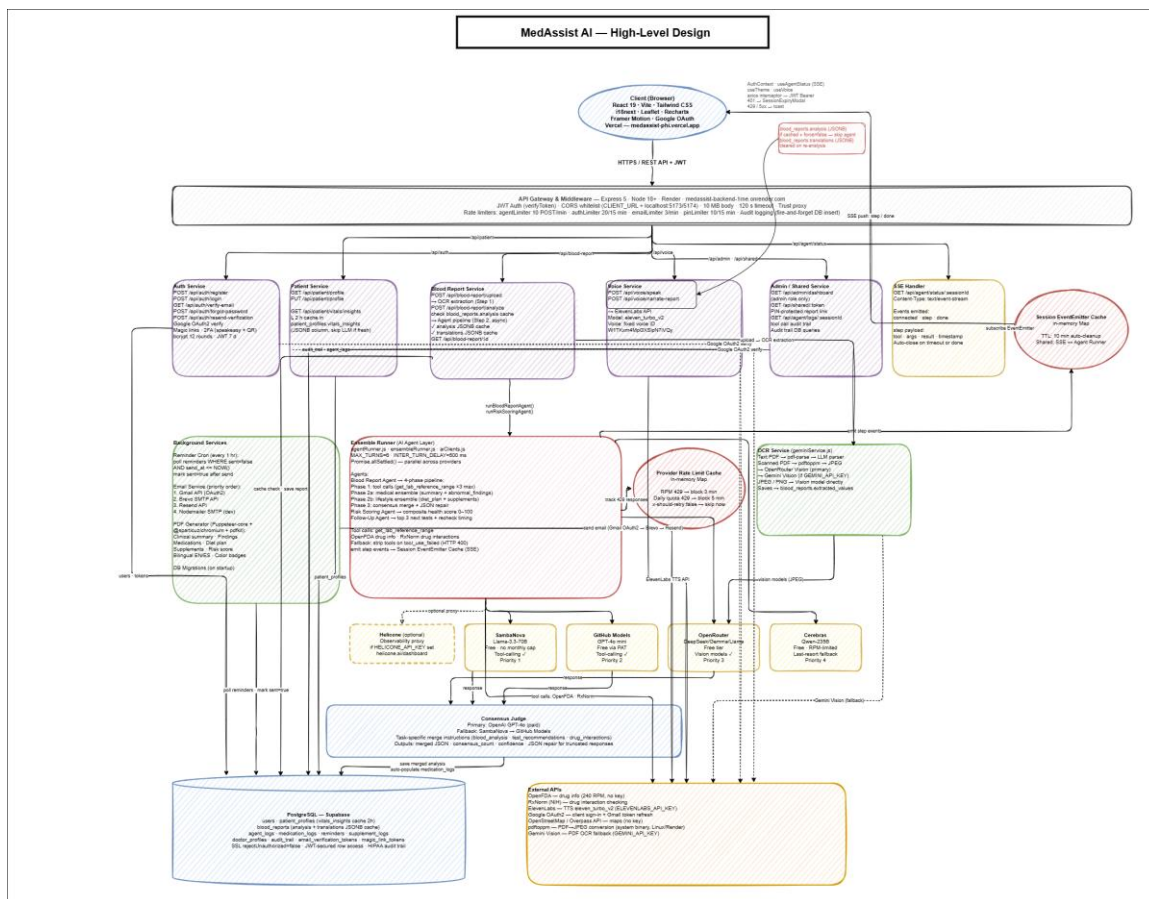
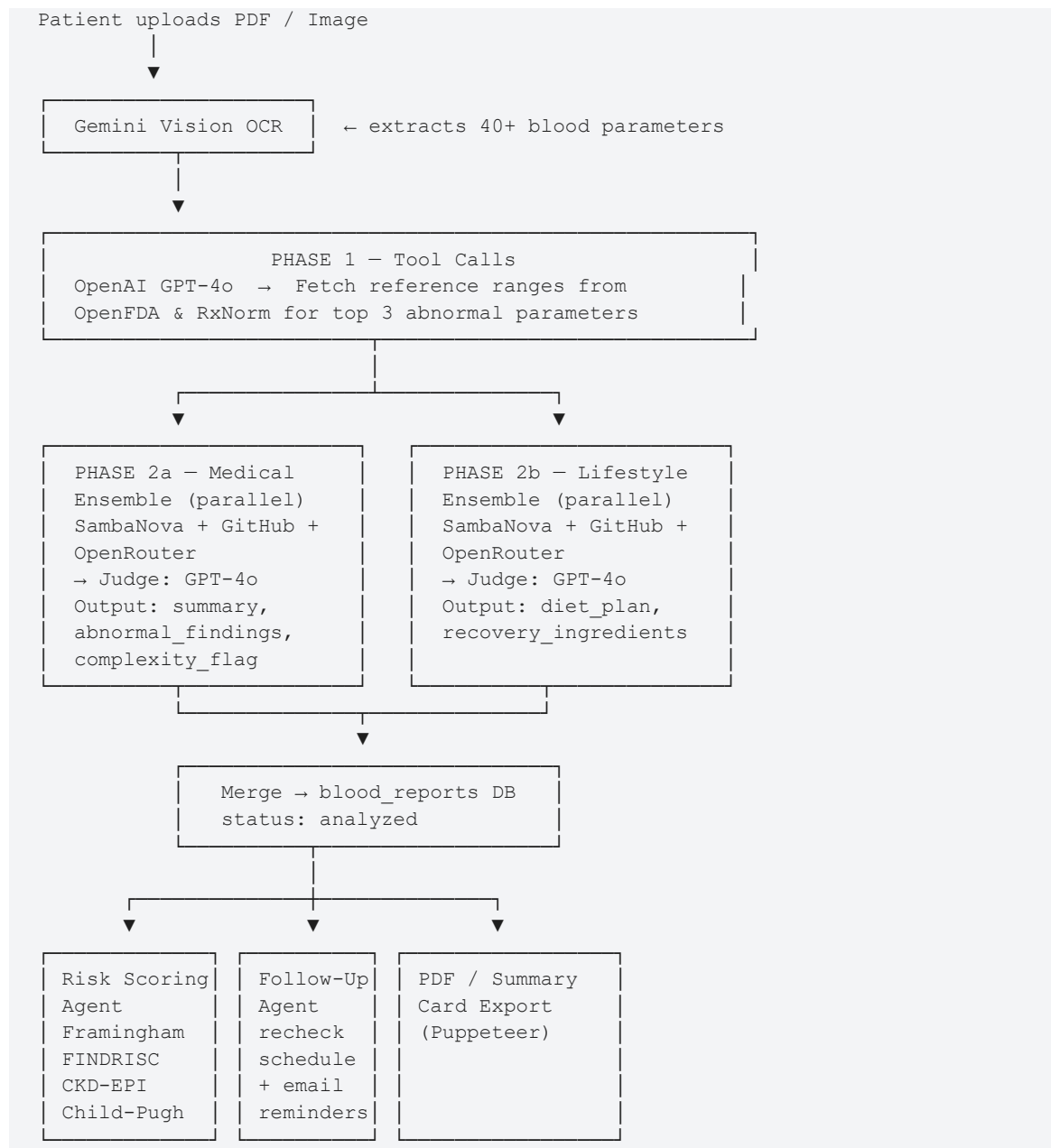


Figure 1 — MedAssist AI System Architecture (HLD)

When a patient uploads a blood report, the system executes a 4-phase pipeline. The result is stored in the database and the patient can immediately view, narrate, export, or share the full analysis.



4. Technology Stack

4.1 Frontend

Technology	Purpose
React 19 + Vite	SPA framework and build tool
Tailwind CSS 3.4	Utility-first styling with dark mode
React Router v7	Client-side routing
Recharts 3.8	Trend charts and sparkline visualizations

Leaflet + React Leaflet	Interactive nearby-clinic map
Framer Motion 12	Page transitions and animations
react-hook-form 7	Form state management
i18next + react-i18next	Full English / Spanish internationalization
Axios 1.13	HTTP client with JWT Bearer interceptor
@react-oauth/google	Google OAuth federated login

4.2 Backend

Technology	Purpose
Node.js 18 + Express 5	REST API server
PostgreSQL via Supabase	Primary relational database
jsonwebtoken	Stateless JWT authentication (7-day tokens)
bcryptjs	Password hashing
Speakeasy + QRCode	TOTP 2-factor authentication
Multer 2.1	File upload handling (10 MB limit)
PDFKit 0.18 + Puppeteer Core	PDF report generation and export
pdf-parse 1.1	Text extraction from digital PDFs
Nodemailer 8	Multi-provider email (Gmail OAuth2, Brevo, Resend)
express-rate-limit 8.3	4-layer rate limiting (auth, email, agent, PIN)

4.3 AI Providers & External APIs

Provider / API	Cost	Role
OpenAI GPT-4o	Paid	Consensus judge + Phase 1 tool calling
SambaNova Llama 3.3 70B	Free	Ensemble agent (parallel inference)
GitHub Models GPT-4o mini	Free*	Ensemble agent (parallel inference)
OpenRouter (multi-chain)	Free	Ensemble fallback agent
Google Gemini 1.5 Vision	Free tier	Blood report OCR — 40+ parameters
ElevenLabs TTS	Paid	Audio narration of findings
OpenFDA	Free	Drug info, adverse events, recall data
RxNorm (NIH)	Free	Drug name normalization and interaction checking
OpenStreetMap Overpass	Free	Nearby clinic/lab/hospital search (10 km)
Google OAuth 2.0	Free	Federated patient authentication

5. Key Features Implemented

5.1 Blood Report Upload

- Upload blood reports as PDF, JPG, or PNG (up to 10 MB)
- Live camera capture mode with guide-frame overlay, front/rear flip, and 2× resolution crop for mobile users
- Google Gemini 1.5 Vision extracts 40+ parameters: name, value, unit, normal range, status
- Graceful handling of multi-page PDFs, handwritten values, and non-standard column layouts

5.2 AI Blood Report Analysis (4-Phase Ensemble)

Output Section	Description
Overall Assessment	2–3 sentence summary with root cause identification
Abnormal Findings	Value, normal range, status (normal/low/high/critical), plain-English interpretation
Complexity Flag	Low / Medium / High with physician referral decision
Diet Plan	Meal schedule, foods to eat/avoid, personalized to the patient's conditions
Recovery Ingredients	Natural supplements with targets and adherence tracking
Tablet Recommendations	FDA-verified medication suggestions with RxNorm normalization

5.3 Clinical Risk Scoring

Four validated clinical models applied to extracted blood values:

- Framingham Risk Score — 10-year cardiovascular disease risk
- FINDRISC — Type 2 diabetes risk (Finnish Diabetes Risk Score)
- CKD-EPI — Estimated GFR for chronic kidney disease staging
- Child-Pugh Score — Liver function and cirrhosis severity
- Composite 0–100 health index with urgency levels: routine / follow-up soon / urgent / emergency

5.4 Follow-Up Scheduling

- Top 3 recommended follow-up tests derived from abnormal finding severity
- Evidence-based recheck timeframes: Critical (1–2 weeks), Significant (1–3 months), Mild (3–6 months)
- Automated background email reminders sent 1 day before each scheduled recheck

5.5 Report History & Trend Analysis

- Full report history with interactive month-by-month timeline
- Recharts trend charts for 10 key parameters across all historical visits
- Side-by-side comparison of any 2 reports with delta badges (↑/↓/→)

5.6 Patient Dashboard

- Composite health score gauge (0–100) with AI summary and risk sparkline
- Daily AI-generated personalized health tips (refreshable, cached 24 hours)
- Recent report cards showing status, parameter count, and abnormal findings
- Achievement badges (first report, analysis complete, streaks)

5.7 Vitals Tracker

Vital	Unit / Notes
Blood Pressure	mmHg — systolic / diastolic
Glucose	mg/dL — fasting or random
Weight	kg
Heart Rate	bpm
SpO ₂	% oxygen saturation
Temperature	°F

30-day trend sparklines per vital type. AI-generated insights correlate vitals to blood report parameters (e.g., elevated glucose correlated with HbA1c result).

5.8 Audio Narration

- Doctor-style full report narration via ElevenLabs TTS (eleven_turbo_v2)
- Plain-English on-demand explanation of any single abnormal finding
- Native Spanish audio narration when language is set to Spanish

5.9 Multi-Language Support (EN / ES)

- Full English / Spanish UI powered by i18next
- Smart translation caching — translations stored in PostgreSQL, incremental key-level updates
- Poisoned-cache detection prevents caching of failed or incomplete translations
- One-click sidebar toggle, persists across sessions

5.10 Report Sharing & PDF Export

- PIN-protected shareable links with 7-day expiry — recipients view the full analysis without an account
- Full analysis PDF export with Puppeteer (headless browser rendering)
- One-page summary card PDF via PDFKit
- Both exports support multi-language output

5.11 Medical ID Card (Emergency)

- Emergency contact name and phone, blood type, organ donor status, critical medical notes
- PIN-protected public lookup by patient ID — accessible to first responders without login

- Brute-force protection: 10 failed PIN attempts triggers 15-minute IP lockout

5.12 Nearby Clinics & Labs

- Searches clinics, labs, hospitals, pharmacies, and blood banks within 10 km of the patient
- Powered by OpenStreetMap Overpass API with 3-mirror parallel fallback (14-second timeout each)
- Results: name, specialization, address, phone, website, and Haversine-calculated distance
- 1-hour in-memory cache per location bucket (up to 200 buckets) — cache hits bypass Overpass entirely
- Circuit breaker: 10-minute cooldown if all mirrors fail simultaneously

5.13 Admin Panel

- System statistics: total users, blood reports uploaded, AI analyses generated
- Paginated user list with search and suspend/unsuspend controls
- HIPAA audit trail: every sensitive action logged with user, action, resource, IP, timestamp

6. AI Agents & Multi-LLM Ensemble

6.1 Blood Report Agent

The primary agent runs a 4-phase pipeline as a fire-and-forget background process. The client polls GET /api/agent/status/:sessionId for real-time progress. Results are persisted to the database immediately as each phase completes.

Phase	Description
Phase 1 — Tool Calls	OpenAI GPT-4o fetches lab reference ranges from OpenFDA and RxNorm for the top 3 abnormal parameters
Phase 2a — Medical Ensemble	SambaNova + GitHub Models run in parallel → GPT-4o merges → summary, abnormal findings, complexity flag, referral decision
Phase 2b — Lifestyle Ensemble	SambaNova + GitHub Models run in parallel → GPT-4o merges → diet plan, recovery ingredients, tablet recommendations
Phase 3 — Persist	Merged JSON written to blood_reports table; status set to 'analyzed'

6.2 Risk Scoring Agent

A single-turn agent that applies four validated clinical scoring algorithms to the extracted blood parameter set combined with the patient's age, weight, BMI, and medical history from their profile.

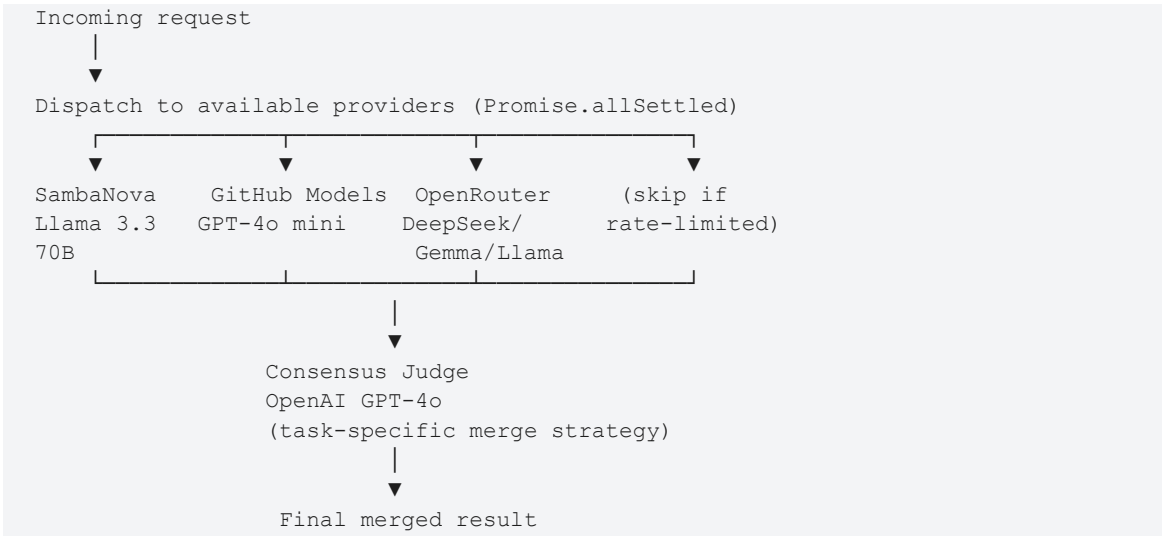
Model	Measures
Framingham	10-year cardiovascular disease probability
FINDRISC	Type 2 diabetes risk (validated in 10+ European cohorts)
CKD-EPI	eGFR estimation for chronic kidney disease staging

Child-Pugh	Liver function and cirrhosis prognosis scoring
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6.3 Follow-Up Agent

Takes the abnormal findings list and the patient's current medications as input. Returns the top 3 recommended follow-up tests with recheck timeframes and urgency levels. Automatically creates a reminder record in the database; the background reminder service sends the email 1 day before each scheduled recheck date.

6.4 Ensemble Runner Architecture



Merge strategies per task type:

Task	Merge Strategy
blood_analysis	Conservative — prefer lower/safer values; flag abnormal if ANY agent raises concern
drug_interactions	Use the MORE SEVERE rating when agents disagree (patient safety priority)
treatment_plan	Prefer safer/lower doses; flag all drug interactions

7. Database Design

MedAssist AI uses PostgreSQL hosted on Supabase. The schema is applied via a base schema.sql file and 4 incremental migrations. Key design choices include JSONB columns for flexible AI output storage and a dedicated audit_trail table for HIPAA compliance.

Table	Purpose & Key Columns
users	Patient and admin accounts — email, password_hash, role, name, 2FA secret

patient_profiles	Health demographics — age, gender, weight, height, conditions, allergies, medications, insurance
blood_reports	Upload + full AI analysis — extracted_values JSONB, analysis JSONB, risk_scores JSONB, follow_up JSONB, translations JSONB
vitals_logs	Daily vital signs — BP systolic/diastolic, glucose, weight, HR, SpO ₂ , temperature
medication_logs	Medication adherence timestamps
supplement_logs	Recovery ingredient adherence and streak counters
medical_id	Emergency card — blood type, organ donor flag, PIN hash, emergency contacts
email_verification_tokens	24-hour registration verification links
password_reset_tokens	24-hour one-time-use password reset tokens
report_shares	PIN-protected 7-day shareable links — token, pin_hash, access_count, expiry
agent_logs	AI agent execution audit — steps JSONB, total_turns, provider, created_at
audit_trail	HIPAA action log — user_id, action, resource, ip_address, user_agent, created_at
reminders	Scheduled follow-up email reminders — send_at, status, test_name, report_id

8. Security & HIPAA Compliance

8.1 Authentication

Mechanism	Details
JWT (7-day tokens)	Stateless auth — payload: userId, role, name; auto-attached by Axios interceptor
Email Verification	Required on registration — 24-hour time-limited link
Password Reset	24-hour time-limited token, one-time use, invalidated after use
Google OAuth 2.0	Federated login — no password stored
TOTP 2FA	Google Authenticator / Authy compatible; QR code setup flow

8.2 Rate Limiting (4 Layers)

Layer	Limit
Auth endpoints	20 requests / 15 min / IP
Email endpoints	3 requests / 60 sec / IP

AI agent endpoints	10 requests / 60 sec / IP
Medical ID PIN lookup	10 attempts / 15 min / IP — brute-force lockout

8.3 HIPAA Audit Trail

Every sensitive action is logged to the `audit_trail` table with: `user_id`, action type, resource identifier, IP address, user-agent string, and timestamp. The admin panel exposes this log with filtering by user, action type, and date range.

9. API Reference

Route Group	Endpoints
/api/auth	register, login, google, forgot-password, verify-email, reset-password, 2fa/setup, 2fa/verify
/api/patient	profile (GET/PUT), vitals (POST/GET), vitals/insights, medical-id, badges, sessions
/api/blood-report	upload, analyze, history, /:id, risk-scores, follow-up, export-pdf, export-summary
/api/voice	speak, narrate-report, explain-finding, translate
/api/shared	share-report, shared/:token, medical-id/:patientId (public PIN lookup)
/api/admin	stats, users (paginated), users/:id/suspend, audit-log
/api/agent	status/:sessionId (polling endpoint for real-time progress)

10. Deployment

Component	Platform	Notes
Frontend	Vercel	React SPA — continuous deploy from main branch
Backend API	Render	Node.js Express server — free tier (ephemeral filesystem)
Database	Supabase	PostgreSQL — SSL required (rejectUnauthorized: false)
CI/CD	GitHub Actions	Lint + build check on every push to main

Live URLs:

Service	URL
Frontend (Vercel)	https://medassist-phi.vercel.app/
Backend API (Render)	https://medassist-backend-1rne.onrender.com

11. Local Setup Guide

11.1 Prerequisites

- Node.js 18+
- PostgreSQL database or a Supabase project
- API keys: OpenAI (required — consensus judge), Gemini (required — OCR), SambaNova + GitHub PAT + OpenRouter (free ensemble agents), ElevenLabs (optional — TTS)
- Google OAuth Client ID and Secret (for federated login)

11.2 Install & Configure

```
# Clone and install
git clone https://github.com/SiddharthBhamare01/medassist-ai.git
cd medassist-ai
cd server && npm install
cd ../client && npm install

# Copy environment files
cp server/.env.example server/.env      # fill in your API keys
cp client/.env.example client/.env      # fill in VITE_API_URL and
VITE_GOOGLE_CLIENT_ID
```

11.3 Database Setup

Run the following SQL files in order in your Supabase SQL editor:

```
server/db/schema.sql
server/db/migrations/001_recommended_tests_jsonb.sql
server/db/migrations/002_session_status.sql
server/db/migrations/003_all_features.sql
server/db/migrations/004_persist_cached_llm_outputs.sql
server/db/seed.sql      # optional demo data
```

11.4 Run Locally

```
# Terminal 1 — backend (http://localhost:5000)
cd server && npm run dev

# Terminal 2 — frontend (http://localhost:5173)
cd client && npm run dev
```

12. Key Design Decisions

Why Multi-Provider Ensemble Instead of a Single Model?

Medical AI requires higher accuracy than general text generation. A single LLM can hallucinate drug names, misstate reference ranges, or produce overconfident risk assessments. By dispatching the same prompt to multiple free-tier providers in parallel and using GPT-4o as a consensus judge, MedAssist achieves higher reliability — similar to the clinical practice of seeking a second opinion. The conservative merge strategy (flag abnormal if any agent raises concern) prioritizes patient safety over false reassurance.

Why Tool Calling Instead of Pure LLM Generation?

Drug names, interaction data, and adverse events must come from authoritative sources — not from LLM training data which can be stale or incorrect. Tool calling in Phase 1 forces the agent to query OpenFDA and RxNorm for real-time, source-verified data before generating any recommendations. This grounds the AI output in facts rather than inference.

Why Fire-and-Forget with Polling Instead of SSE?

The 4-phase blood report pipeline can take 30–90 seconds. Keeping an HTTP connection open that long is unreliable on serverless/free-tier hosting (Render sleeps idle connections). Fire-and-forget stores the job, and the client polls GET /api/agent/status/:sessionId every 3 seconds. The AgentStatusPanel renders real-time progress bars so the wait feels transparent and trustworthy.

Why Gemini Vision for OCR?

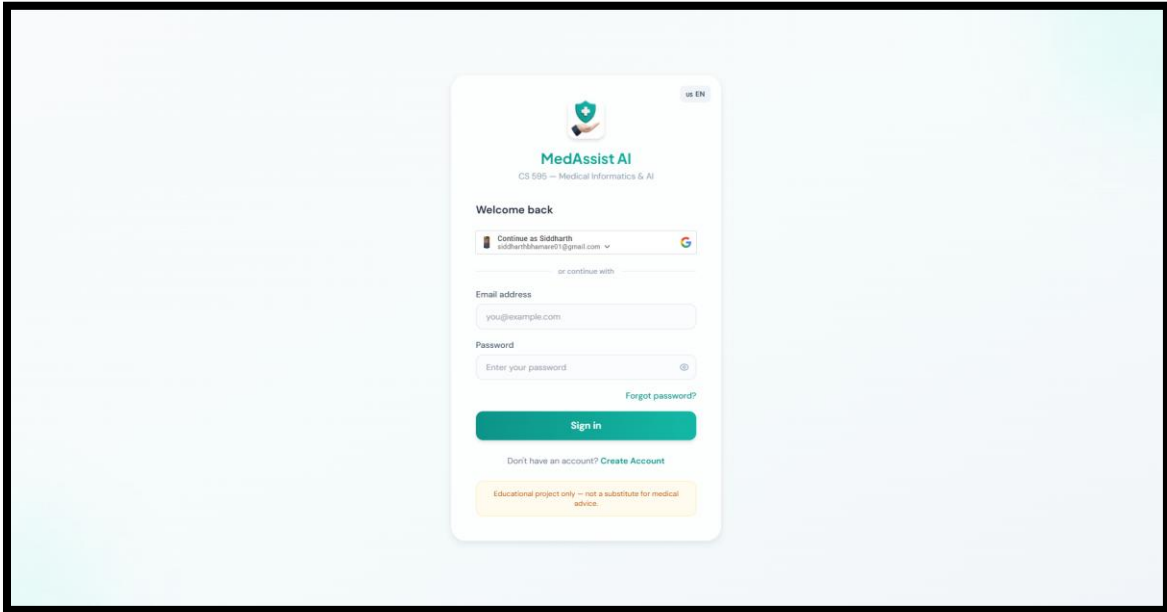
Blood reports come in dozens of formats — scanned images, photographed pages, multi-column PDFs, handwritten annotations. Traditional pdf-parse works for digital PDFs but fails on images. Gemini 1.5 Vision handles both with a structured JSON prompt describing the 40+ expected parameters, gracefully returning null for any value it cannot confidently extract rather than hallucinating a value.

13. Educational Disclaimer

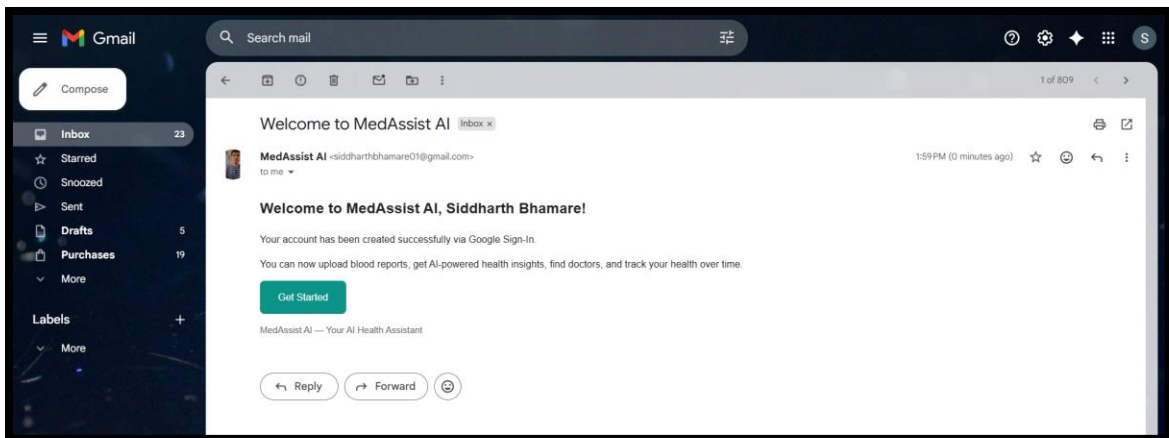
MedAssist AI is an educational CS 595 project developed at Illinois Institute of Technology. It is not a licensed medical device and must not be used for actual clinical decisions. All AI-generated outputs are labeled as educational information only. The system never prescribes medications — it suggests informational references. Complex or critical findings are flagged for physician consultation. All system prompts explicitly include the disclaimer: "Educational use only — not a substitute for professional medical advice."

Screenshots

Authentication & Onboarding

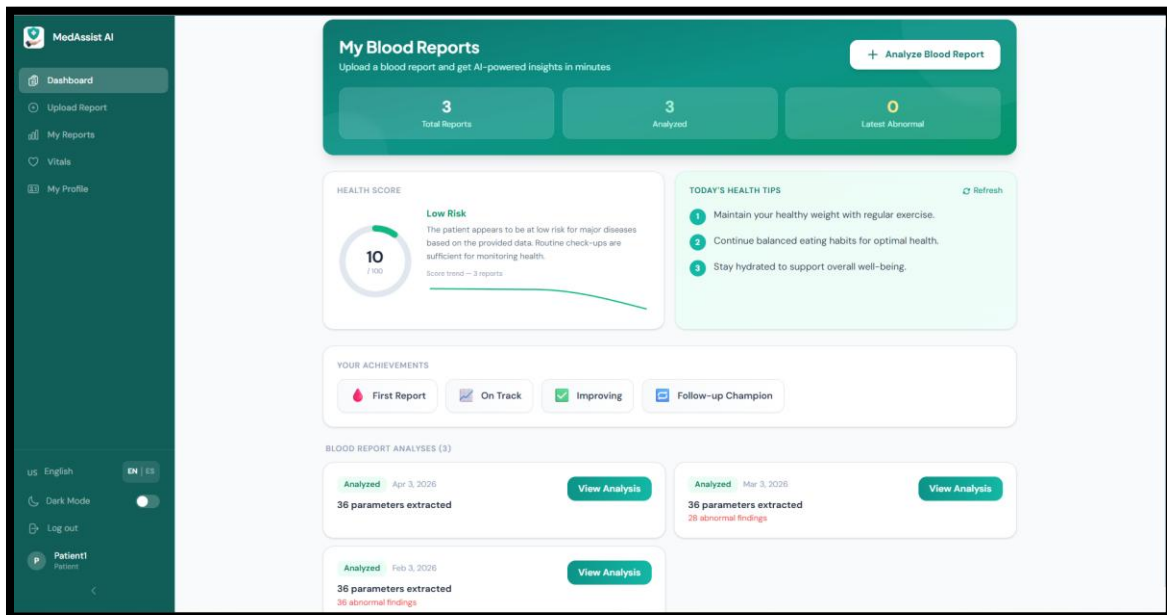


Login Page with Google OAuth



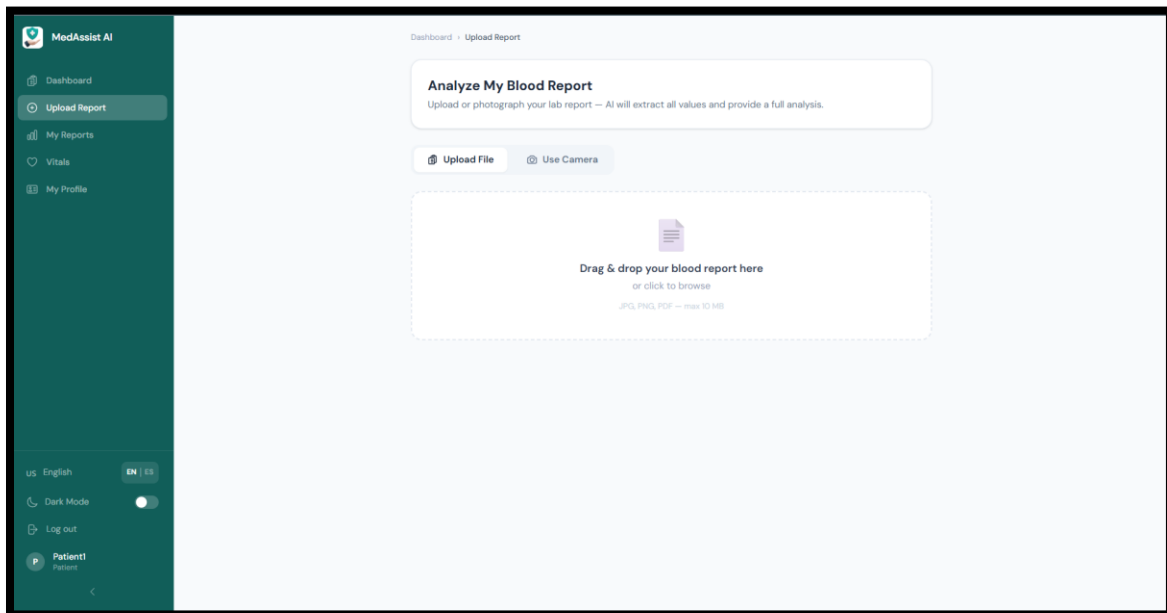
Welcome Email on New Account Creation

Patient Dashboard



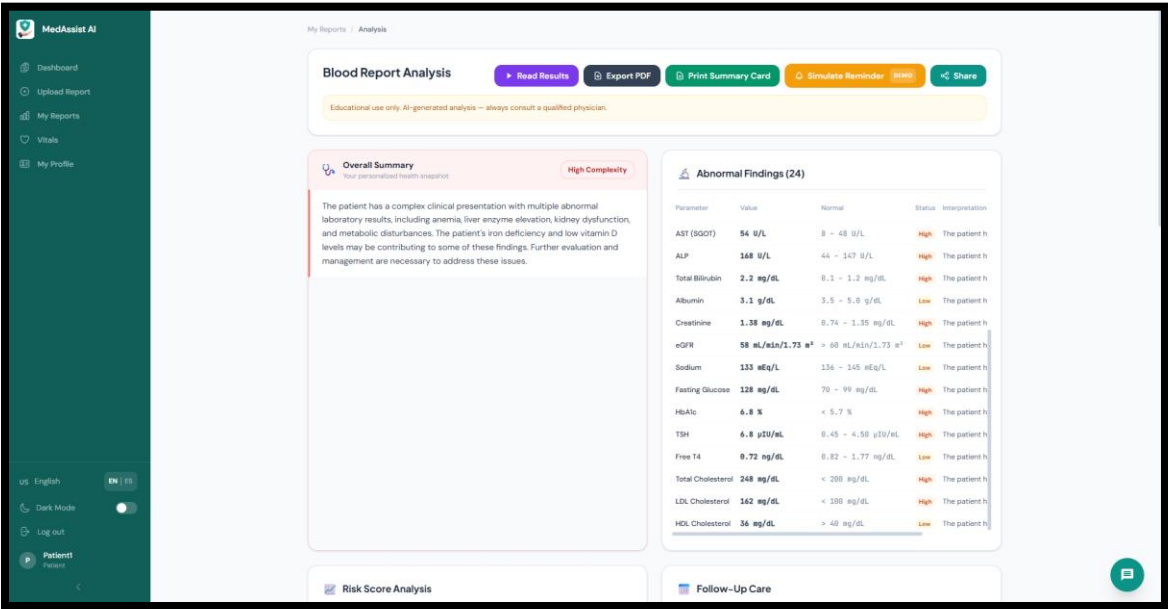
Patient Dashboard — Health Score, Tips & Report Cards

Blood Report Upload

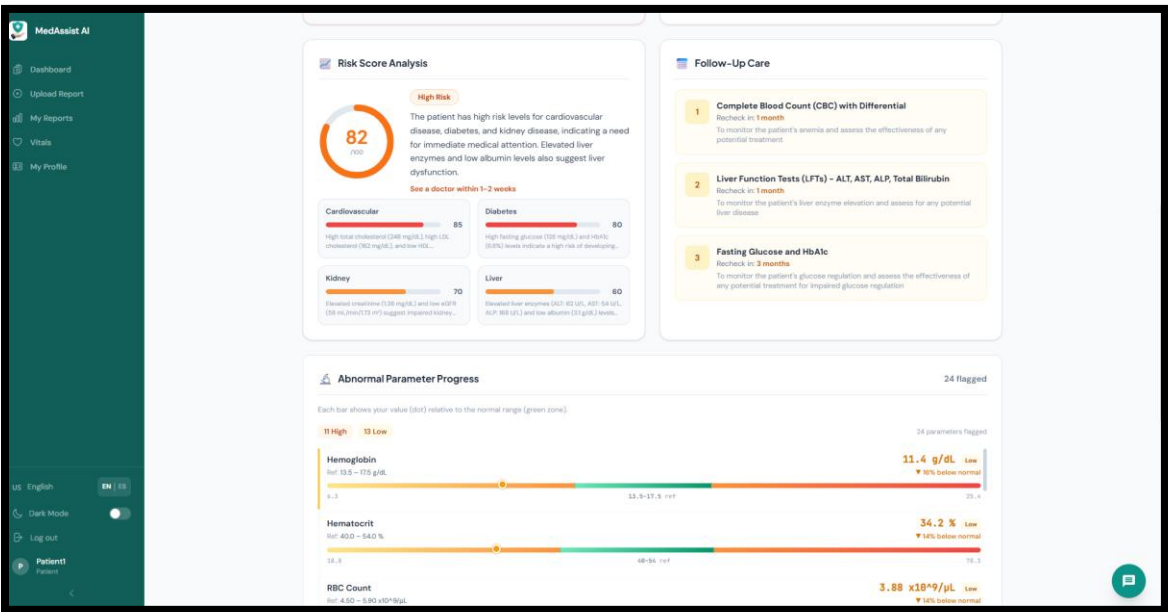


Blood Report Upload (File or Camera)

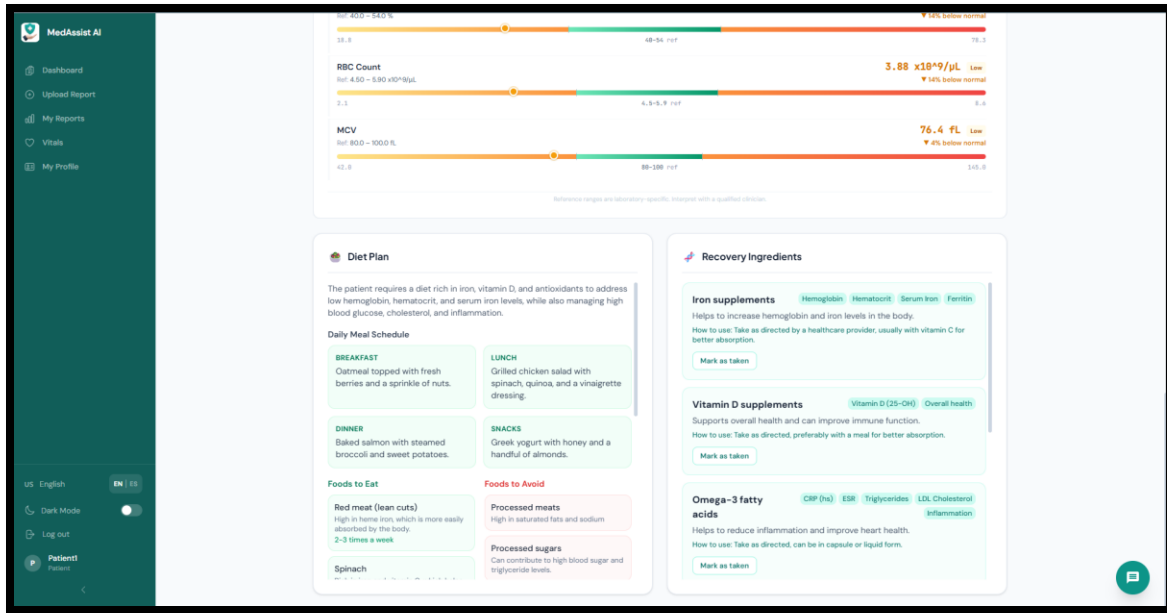
AI Blood Report Analysis



Analysis — Overall Summary & Abnormal Findings

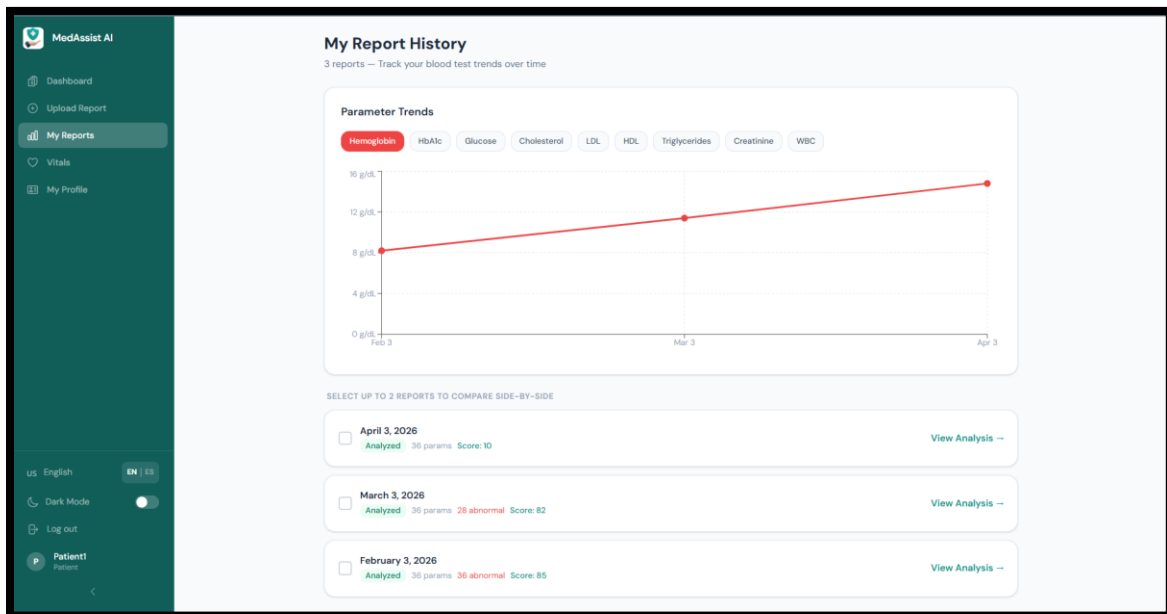


Analysis — Risk Score Breakdown & Follow-Up Care

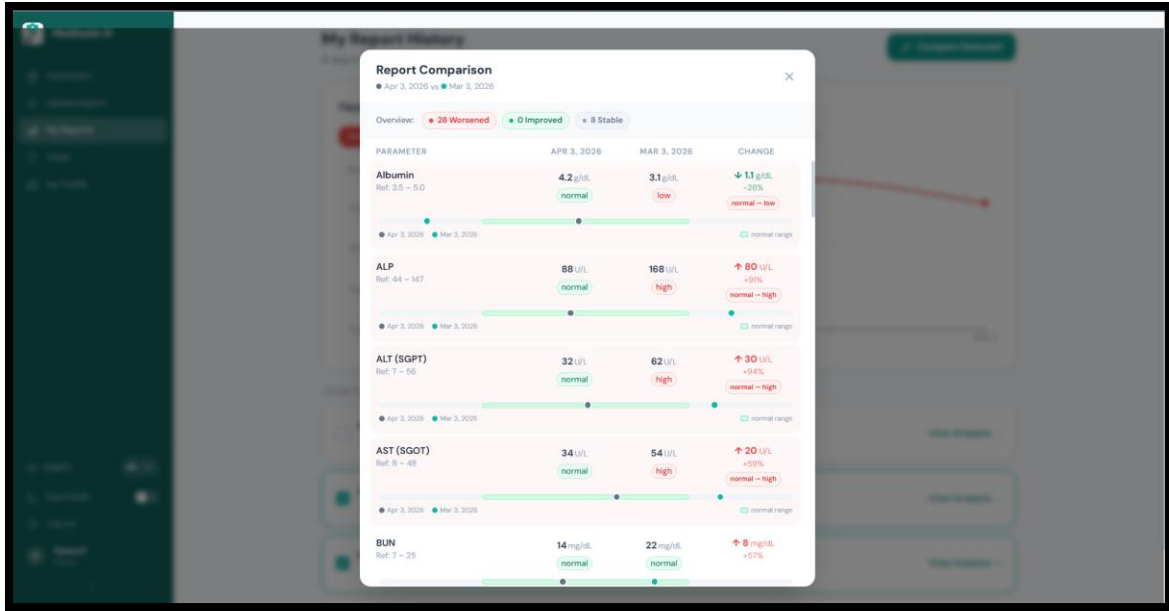


Analysis — Diet Plan & Recovery Ingredients

Report History & Comparison

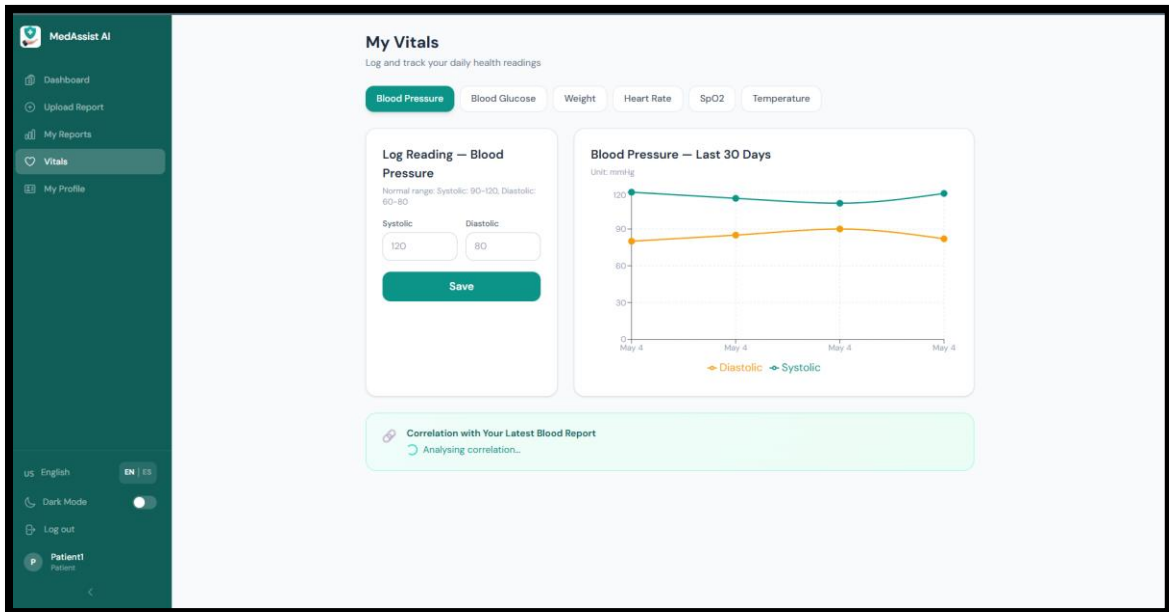


My Report History with Parameter Trend Charts



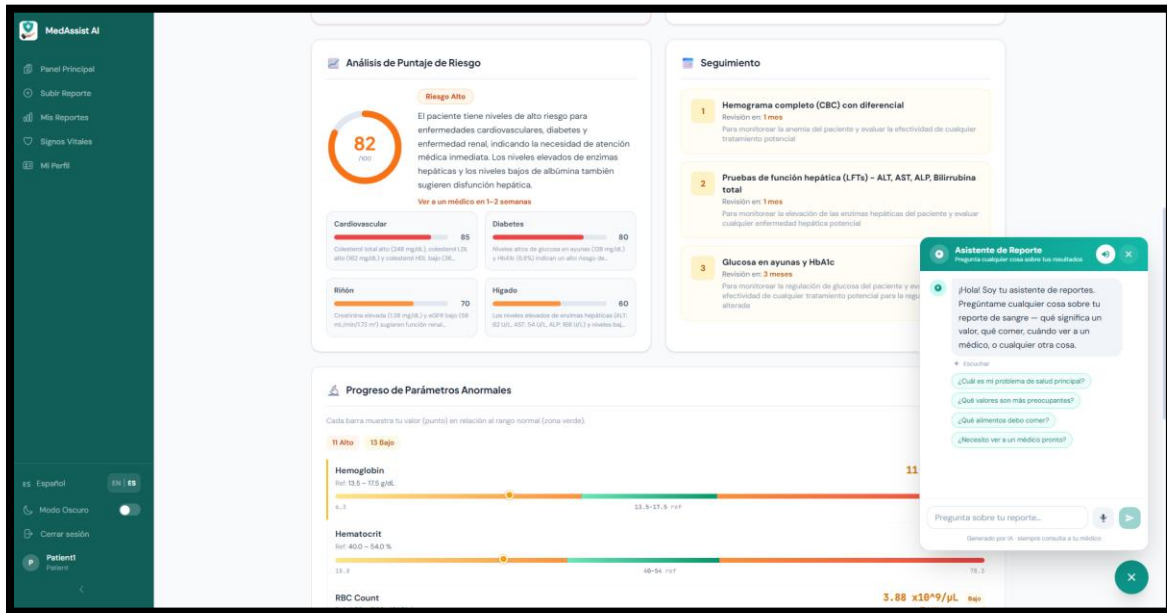
Side-by-Side Report Comparison with Delta Badges

Vitals Tracker



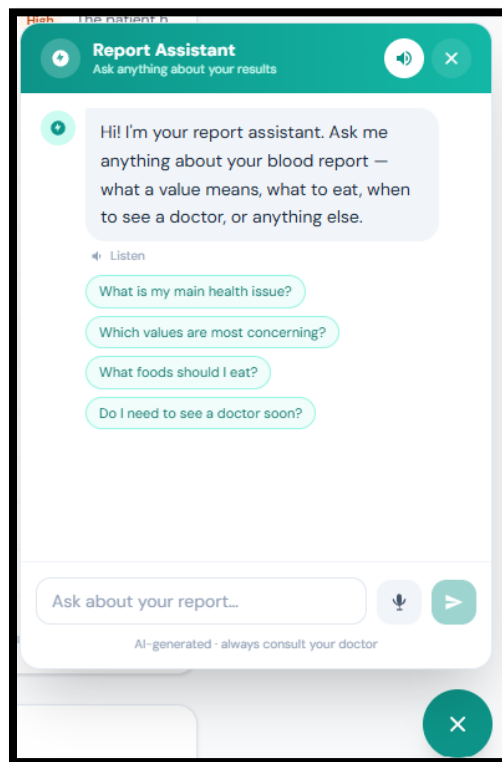
Vitals Tracker — Blood Pressure 30-Day Trend

Multi-Language Support



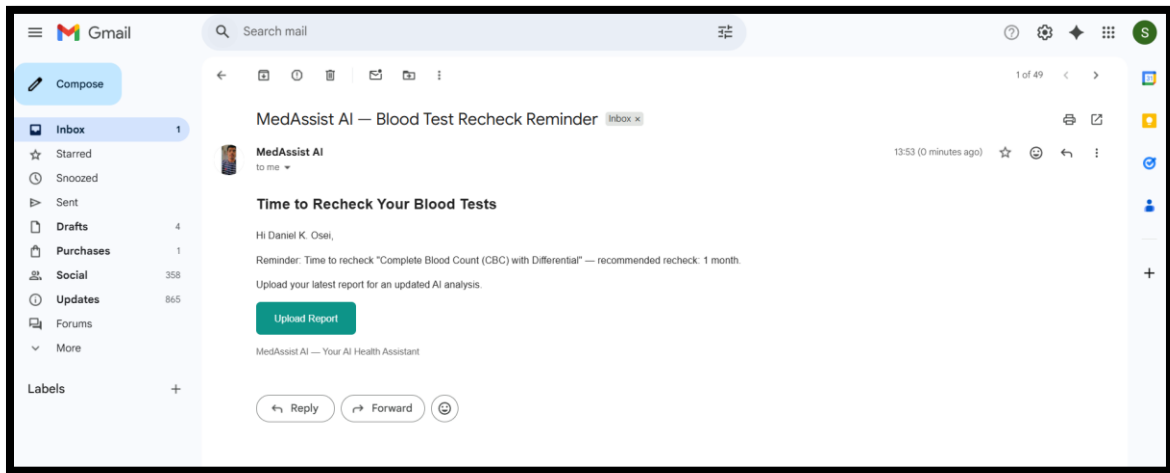
Full Spanish UI with AI Chatbot Assistant

AI Chatbot Assistant



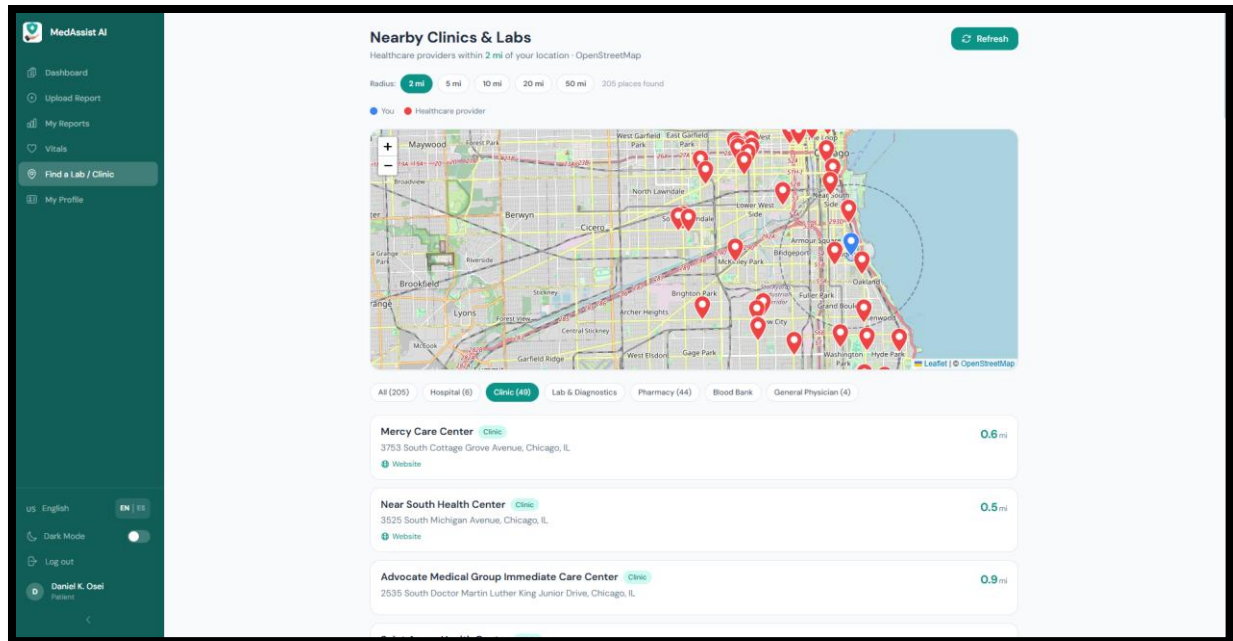
Report Assistant Chatbot (English)

Follow-Up Email Reminder



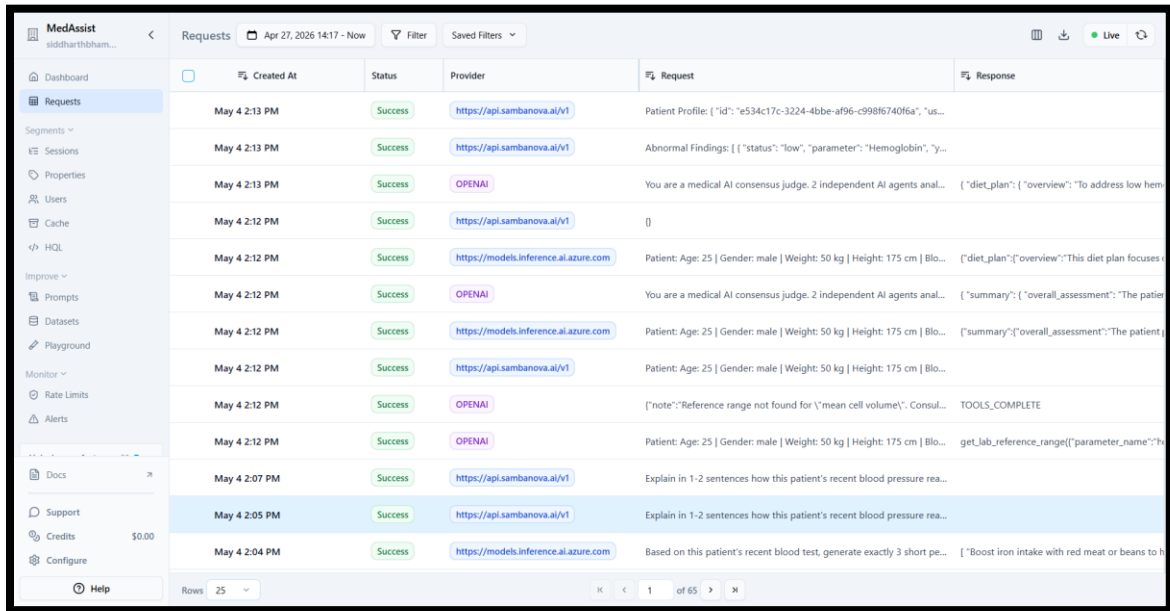
Automated Blood Test Recheck Reminder Email

Suggestion of Clinics/lab as per location of Patient



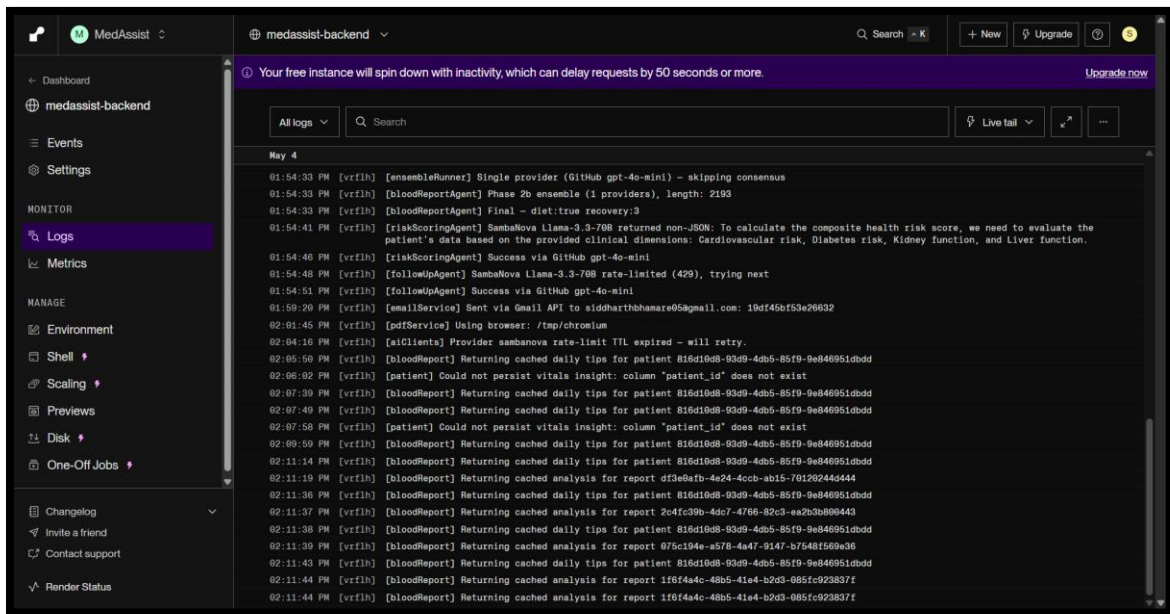
Location based clinic/lab recommendation

Live Agent Tracking & Backend Logs



Created At	Status	Provider	Request	Response
May 4 2:13 PM	Success	https://api.sambanova.ai/v1	Patient Profile: {"id": "e534c17c-3224-4bbe-af96-c998f6740f6a", "us...	
May 4 2:13 PM	Success	https://api.sambanova.ai/v1	Abnormal Findings: [{"status": "low", "parameter": "Hemoglobin", "y...	
May 4 2:13 PM	Success	OPENAI	You are a medical AI consensus judge. 2 independent AI agents anal...	{"diet_plan": {"overview": "To address low hem...
May 4 2:12 PM	Success	https://api.sambanova.ai/v1		
May 4 2:12 PM	Success	https://models.inference.ai.azure.com	Patient: Age: 25 Gender: male Weight: 50 kg Height: 175 cm Blo...	{"diet_plan": {"overview": "This diet plan focuses i...
May 4 2:12 PM	Success	OPENAI	You are a medical AI consensus judge. 2 independent AI agents anal...	{"summary": {"overall_assessment": "The patie...
May 4 2:12 PM	Success	https://models.inference.ai.azure.com	Patient: Age: 25 Gender: male Weight: 50 kg Height: 175 cm Blo...	{"summary": {"overall_assessment": "The patient...
May 4 2:12 PM	Success	https://api.sambanova.ai/v1	Patient: Age: 25 Gender: male Weight: 50 kg Height: 175 cm Blo...	
May 4 2:12 PM	Success	OPENAI	["note": "Reference range not found for 'mean cell volume'", "Consul...	TOOLS_COMPLETE
May 4 2:12 PM	Success	OPENAI	Patient: Age: 25 Gender: male Weight: 50 kg Height: 175 cm Blo...	get_lab_reference_range(["parameter_name": "h...
May 4 2:07 PM	Success	https://api.sambanova.ai/v1	Explain in 1-2 sentences how this patient's recent blood pressure rea...	
May 4 2:05 PM	Success	https://api.sambanova.ai/v1	Explain in 1-2 sentences how this patient's recent blood pressure rea...	
May 4 2:04 PM	Success	https://models.inference.ai.azure.com	Based on this patient's recent blood test, generate exactly 3 short pe...	["Boost iron intake with red meat or beans to h...

Live Agent API Call Tracking (Langfuse)



Timestamp	Log Message
01:54:33 PM [vrf:h]	[ensembleRunner] Single provider (GitHub gpt-4o-mini) - skipping consensus
01:54:33 PM [vrf:h]	[bloodReportAgent] Phase 2b ensemble (1 providers), length: 2193
01:54:33 PM [vrf:h]	[bloodReportAgent] Final - diet:true recovery:3
01:54:41 PM [vrf:h]	[riskScoringAgent] SambaNova Llama-3.3-70B returned non-JSON: To calculate the composite health risk score, we need to evaluate the patient's data based on the provided clinical dimensions: Cardiovascular risk, Diabetes risk, Kidney function, and Liver function.
01:54:40 PM [vrf:h]	[riskScoringAgent] Success via GitHub gpt-4o-mini
01:54:48 PM [vrf:h]	[followUpAgent] SambaNova Llama-3.3-70B rate-limited (429), trying next
01:54:51 PM [vrf:h]	[followUpAgent] Success via GitHub gpt-4o-mini
01:59:20 PM [vrf:h]	[emailService] Sent via Gmail API to siddharthbhamare95@gmail.com: 19df45bf53a26632
02:01:45 PM [vrf:h]	[pdfService] Using browser: /tmp/chromium
02:04:16 PM [vrf:h]	[aiClients] Provider sambanova rate-limit TTL expired - will retry.
02:06:50 PM [vrf:h]	[bloodReport] Returning cached daily tips for patient 816d19d8-93d9-4db5-85f9-9e846951dbdd
02:06:02 PM [vrf:h]	[patient] Could not persist vitals insight: column "patient_id" does not exist
02:07:39 PM [vrf:h]	[bloodReport] Returning cached daily tips for patient 816d19d8-93d9-4db5-85f9-9e846951dbdd
02:07:49 PM [vrf:h]	[bloodReport] Returning cached daily tips for patient 816d19d8-93d9-4db5-85f9-9e846951dbdd
02:07:58 PM [vrf:h]	[patient] Could not persist vitals insight: column "patient_id" does not exist
02:09:58 PM [vrf:h]	[bloodReport] Returning cached daily tips for patient 816d19d8-93d9-4db5-85f9-9e846951dbdd
02:11:14 PM [vrf:h]	[bloodReport] Returning cached daily tips for patient 816d19d8-93d9-4db5-85f9-9e846951dbdd
02:11:19 PM [vrf:h]	[bloodReport] Returning cached analysis for report df3e9afb-4e24-4ccb-ab15-70120744a444
02:11:36 PM [vrf:h]	[bloodReport] Returning cached daily tips for patient 816d19d8-93d9-4db5-85f9-9e846951dbdd
02:11:37 PM [vrf:h]	[bloodReport] Returning cached analysis for report 2e4fc39b-4dc7-4766-82c3-aa2b3b080443
02:11:38 PM [vrf:h]	[bloodReport] Returning cached daily tips for patient 816d19d8-93d9-4db5-85f9-9e846951dbdd
02:11:39 PM [vrf:h]	[bloodReport] Returning cached analysis for report 075c194e-a578-4a47-9147-b7548f569e36
02:11:43 PM [vrf:h]	[bloodReport] Returning cached daily tips for patient 816d19d8-93d9-4db5-85f9-9e846951dbdd
02:11:44 PM [vrf:h]	[bloodReport] Returning cached analysis for report 1f6f4a4c-48b5-41e4-b2d3-085fc923837f
02:11:44 PM [vrf:h]	[bloodReport] Returning cached analysis for report 1f6f4a4c-48b5-41e4-b2d3-085fc923837f

Live Backend Server Logs on Render